MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – PERIODIC TEST 10** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 9: Some Applications of Trigonometry** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Case Study Question 12(iii).
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 2 Very Short Answer questions of 2 marks each.
4. **Section C** comprises 2 Short Answer questions of 3 marks each.
5. **Section D** comprises 2 Long Answer / HOTS questions of 5 marks each.
6. **Section E** comprises 1 Applied Case Study question of 5 marks with internal choice.
7. Use of calculators is not permitted. Take $\sqrt{3} = 1.732$ unless stated otherwise.

SECTION A: Multiple Choice & Assertion-Reason (5 × 1 = 5 Marks)

Q1. A vertical stick 20 cm long casts a shadow 15 cm long on the ground. At the same time, a tower casts a shadow 45 m long on the ground. The height of the tower is: [1]

(A) 40 m (B) 50 m (C) 60 m (D) 75 m

Q2. A kite is flying with a string of length 100 m. If the string makes an angle of 30° with the level ground and there is no sag, the height of the kite is: [1]

(A) 50 m (B) $50\sqrt{3}$ m (C) $\dfrac{100}{\sqrt{3}}$ m (D) 25 m

Q3. If the length of the shadow of a vertical pole decreases, then the angle of elevation of the sun: [1]

(A) increases (B) decreases (C) remains constant (D) first decreases then increases

Q4. A bridge across a river makes an angle of 45° with the river bank. If the length of the bridge across the river is 150 m, the perpendicular width of the river is: [1]

(A) $75\sqrt{2}$ m (B) 75 m (C) $150\sqrt{2}$ m (D) $\dfrac{150}{\sqrt{3}}$ m

Q5. **Assertion (A):** An observer of height 1.5 m standing 20 m from a vertical pole sees the top of the pole at an angle of elevation of 45°. The total height of the pole is 21.5 m.
Reason (R): Total height of an object is obtained by adding the height of the observer to the vertical height above eye level: $H = h_{\text{observer}} + d\tan\theta$. [1]

(A) Both (A) and (R) are true; (R) explains (A).
(B) Both are true; (R) does not explain (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

SECTION B: Very Short Answer Questions (2 × 2 = 4 Marks)

Q6. A bridge over a canal makes an angle of 45° with the canal bank. If the length of the bridge is 100 m, find the perpendicular width of the canal. [2]

Q7. Find the angle of elevation of the sun when the shadow of a pole h metres high is $\sqrt{3}h$ metres long. [2]

SECTION C: Short Answer Questions (2 × 3 = 6 Marks)

Q8. The angles of depression of the top and bottom of a 50 m high building from the top of a tower are observed to be 30° and 60° respectively. Find the height of the tower and the horizontal distance between them. [3]

Q9. The shadow of a tower is found to be 60 m shorter when the sun's altitude changes from 30° to 45°. Find the height of the tower. [Take $\sqrt{3} = 1.732$] [3]

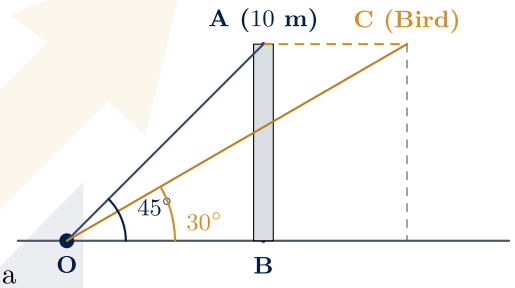
SECTION D: Long Answer / HOTS Questions (2 × 5 = 10 Marks)

Q10. An aeroplane flying at an altitude of 1200 m observes two ships on the same side moving away from it in the same vertical plane. The angles of depression of the ships are observed to be 60° and 30° respectively. Find the distance between the two ships. [Take $\sqrt{3} = 1.732$] [5]

Q11. The angle of elevation of a stationary cloud from a point 20 m above the surface of a lake is 30° and the angle of depression of its reflection in the lake is 60°. Find the height of the cloud above the surface of the lake. [5]

SECTION E: Applied Case Study (1 × 5 = 5 Marks)

Q12. CASE STUDY: Residential Garden Avian Flight Dynamics



In a landscaped public park, a migratory bird is perched at the top of a tall mango tree AB of height 10 m. A student seated on a park bench at point O on level ground observes the bird at an angle of elevation of 45°.

Suddenly, the bird takes off and flies away horizontally in a straight line at a constant altitude of 10 m. Exactly 2 seconds later, the student observes the bird at position C at an angle of elevation of 30°.

Based on the above mathematical scenario, answer the following questions:

(i) Find the initial horizontal distance of the tree from the observer (OB). [1]

(ii) Find the horizontal distance of the bird from the observer after 2 seconds (OD). [2]

(iii) Calculate the flight speed of the bird in m/s and km/h. [Take $\sqrt{3} = 1.732$] [2]

— OR —

If the bird had instead flown towards the observer until its angle of elevation became 60°, find its new horizontal distance from the observer. [2]

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